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Q1740 Starts with $\theta = 0$, when $\theta = 30^\circ$,

$v_A, v_B = ?$

$$\Delta y_A = \frac{L}{2} \sin \theta$$

$$= \frac{L}{2} \sin \theta$$

B 3B 3410) $S_B = L \sin \theta$

$$\Delta y_B = S_B \sin \theta$$

$$\Delta y_A = \frac{1}{2} \Delta y_B = \frac{1}{2} S_B \sin \theta$$

$$E_0 = 0 = E_{\theta=30}$$

$$0 = -mg \Delta y_A + (T)$$

$$\therefore T = \frac{1}{2} m v_A^2 + \frac{1}{2} I_G \omega_G^2$$

$$I_G = \frac{1}{12} m L^2$$

$$\therefore mg \frac{\sqrt{3}}{4} S_B = \frac{1}{2} m v_A^2 + \frac{1}{2} \left(\frac{1}{12} m L^2 \right) \omega_G^2$$

$$\therefore v_A = \omega_G \left(\frac{\sqrt{3}}{4} L \right)$$

$$\theta = 30^\circ,$$

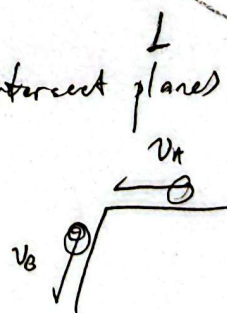
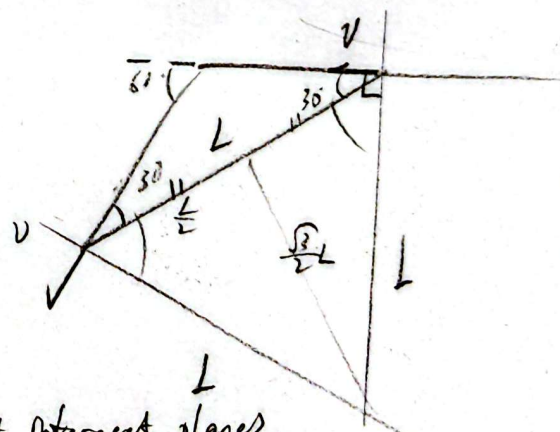
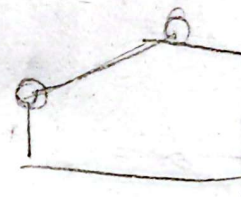
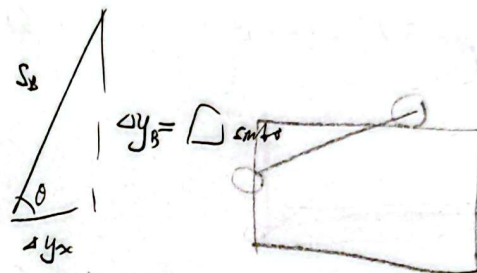
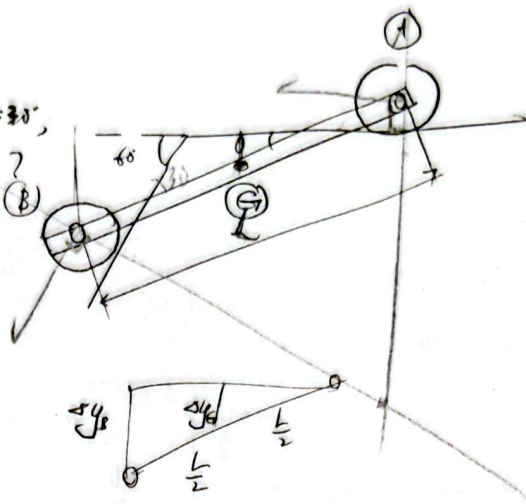
$$\frac{\sqrt{3}}{4} g \left(\frac{L}{2} \right) = \frac{3}{8} \omega_G^2 + \frac{1}{24} L^2 \omega_G^2$$

$$gL = \left(3 + \frac{L^2}{3} \right) \omega_G^2$$

$$\omega_G = \sqrt{\frac{3gL}{9+L^2}}$$

$$\therefore |v_A| = |v_B| = \left(\omega_G L \right) = \sqrt{\frac{3gL^3}{9+L^2}}$$

c direction: ~~accompany~~ to the current intersect planes.
parallel



Q17.89

Angular Momentum Conservation
Energy

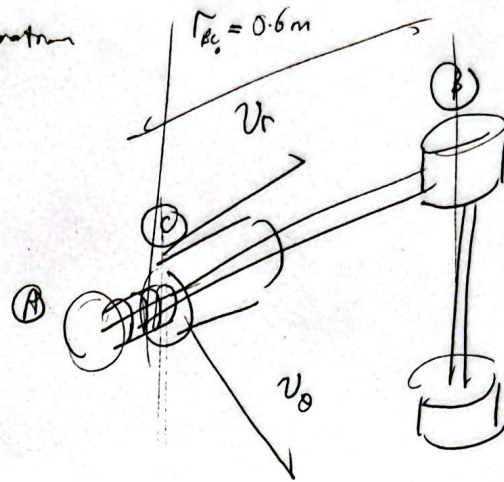
$$\begin{cases} L_{B0}' = L_{B0} \text{ --- (1)} \\ E_{B0}' = E_{B0} \text{ --- (2)} \end{cases}$$

$$\textcircled{1} \quad \therefore \vec{L} = \vec{r} \times m\vec{v} + \dots$$

$$m r_{Bcm}^2 \omega' + I_B \omega' = m r_{B0}^2 \omega + I_B \omega$$

$$(8 r_{Bcm}^2 + 1.2) \omega' = (8 (0.6)^2 (1.5) + (1.2) (1.5))$$

$$(8 r_{Bcm}^2 + 1.2) \omega' = 6.12 \text{ --- (1)}$$



Q

$$T_1' = T_0 \quad \therefore T = \frac{1}{2} m v^2 + \frac{1}{2} I_B \omega^2$$

$$\frac{1}{2} (8) (\omega' r_{Bcm})^2 + \frac{1}{2} (1.2) (\omega')^2 = \frac{1}{2} (8) (1.5 \cdot 0.6)^2 + \frac{1}{2} (1.2) (1.5)^2 + \frac{1}{2} m v_1^2$$

$$\frac{1}{2} (8 r_{Bcm}^2 + 1.2) (\omega')^2 = 3.6 + 1.35 + \frac{1}{2} m v_1^2$$

$$\frac{1}{2} (8) (\omega' r_{Bcm})^2 + \frac{1}{2} (1.2) (\omega')^2 = \frac{1}{2} (8) (v_{0r})^2 + \frac{1}{2} (8) (v_{0t})^2 + \frac{1}{2} I_B (\omega')^2$$

$$\frac{1}{2} (8 r_{Bcm}^2 + 1.2) (\omega')^2 = 13.5 \text{ --- (2)}$$

$$\begin{aligned} r_{cm} &\approx 0.33 \text{ m} \\ \omega &\approx 2.45 \text{ rad/s} \end{aligned}$$

